Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2005-II-Q01] | Part A (Q1-Q5)

Answer: 013

1. The equation $\binom{n}{6} = 6\binom{n}{3}$ simplifies to $(n-3)(n-4)(n-5) = 720$.

2. The product increases for $n \geq 6$, and $10 \cdot 9 \cdot 8 = 720$, so the unique solution is $n = 13$.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2012-II-Q03] | Part A (Q1-Q5)

Answer: 088

1. If every department contributes one man and one woman, there are $4^3 = 64$ committees.

2. Otherwise the gender splits by department must be two men, two women, and mixed. Choose those department roles in $3 \cdot 2$ ways and the mixed pair in four ways, adding twenty-four.

3. The total is 88.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2017-I-Q01] | Part A (Q1-Q5)

Answer: 390

1. Choose any three of15 points, giving $\binom{15}{3} = 455$ triples. A line not coinciding with a triangle side meets the convex boundary in at most two points, so every degenerate triple lies on one side.

2. Including endpoints, the side counts are $5, 6, 7$. Their collinear triples are disjoint and total $\binom{5}{3} + \binom{6}{3} + \binom{7}{3} = 65$. Thus the answer is $455 - 65 = 390$.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2000-II-Q07] | Part B (Q6-Q10)

Answer: 137

1. Multiplying by $19!$ gives $19N = \sum_{k=2}^{9} \binom{19}{k}$. By symmetry the sum from $k = 0$ to 9 is half of 2^{19}, namely 2^{18}.

2. Subtract $\binom{19}{0} + \binom{19}{1} = 20$ to get $19N = 262124$, so $N = 13796$. The requested integer is 137.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2002-II-Q09] | Part B (Q6-Q10)

Answer: 501

1. For an ordered disjoint pair, every element is in the first subset, the second, or neither. There are 3^{10} assignments.

2. Exclude empty first and empty second subsets, giving $3^{10} - 2 \cdot 2^{10} + 1 = 57002$. The two nonempty disjoint subsets cannot be equal, so divide by two: $N = 28501$. Remainder 501.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2007-I-Q10] | Part B (Q6-Q10)

Answer: 860

1. Choose the first column in $\binom{6}{3} = 20$ ways. If the second overlaps it in j rows, its choices number $\binom{3}{j}\binom{3}{3-j}$. There are then j completed rows, j empty rows and $6-2j$ singly shaded rows.

2. The last two columns must both use the empty rows, and column three chooses $3-j$ of the singly shaded rows; the final column is forced. Thus $N = 20\sum_{j=0}^{3}\binom{3}{j}^2\binom{6-2j}{3-j} = 20(20+54+18+1) = 1860$.

3. The remainder is 860.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2009-II-Q06] | Part B (Q6-Q10)

Answer: 750

1. For the complement, write selected numbers $a_1 < \cdots < a_5$ with gaps at least two. Set $b_i = a_i - (i-1)$, a bijection to ordinary five-subsets of $\{1, \ldots, 10\}$.

2. Thus $m = \binom{14}{5} - \binom{10}{5} = 2002 - 252 = 1750$, remainder 750.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2010-I-Q07] | Part B (Q6-Q10)

Answer: 760

1. The unique elements in $A \cap B, B \cap C, C \cap A$ must be distinct. Assign them in $7 \cdot 6 \cdot 5 = 210$ ways.

2. Each of the four remaining elements may belong only to A, only to B, only to C, or none. These 4^4 choices introduce no new intersections.

3. The count is $210 \cdot 256 = 53760$, with remainder 760.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2010-II-Q08] | Part B (Q6-Q10)

Answer: 772

1. For $|A| = k$, $1 \le k \le 11$, the conditions force $k \in B$ and $12 - k \in A$. At $k = 6$ this is impossible.

2. For every other k, choose the other $k - 1$ elements of A from ten unrestricted elements: $\binom{10}{k-1}$. The complement fixes B.

3. The sum is $\sum_{j=0}^{10} \binom{10}{j} - \binom{10}{5} = 1024 - 252 = 772$.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2017-I-Q07] | Part B (Q6-Q10)

Answer: 564

1. Set $c = 6 - a - b$ and use binomial symmetry. The sum becomes $\sum_{a+b+c=6} \binom{6}{a}\binom{6}{b}\binom{6}{c}$.

2. This counts choosing six objects from three disjoint six-object groups, classified by the number chosen from each. Thus $S = \binom{18}{6} = 18564$, with remainder564.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2020-I-Q07] | Part B (Q6-Q10)

Answer: 081

1. Choosing k men and $k+1$ women gives $N=\sum_{k=0}^{11}\binom{11}{k}\binom{12}{11-k}=\binom{23}{11}$. The second equality counts eleven selections from two disjoint groups.

2. In $23!/(11!12!)$, the nonzero prime exponents are those of $2,7,13,17,19,23$, each equal to one. For example $v_2=19-8-10=1$, while $v_3=9-4-5=0$.

3. The sum of these primes is $2+7+13+17+19+23=81$.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2021-II-Q06] | Part B (Q6-Q10)

Answer: 454

1. Let $a = |A|, b = |B|, i = |A \cap B|$. The equation factors as $(a - i)(b - i) = 0$, so one set contains the other.

2. For $A \subseteq B$, each element is in neither, in B only, or in both: 3^5 choices. Reverse containment contributes another 3^5, while equality contributes 2^5 to both. The answer is $2 \cdot 3^5 - 2^5 = 454$.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2022-II-Q10] | Part B (Q6-Q10)

Answer: 004

1. Expand the nested binomial: $\binom{\binom{k}{2}}{2} = (k+1)k(k-1)(k-2)/8 = 3\binom{k+1}{4}$.

2. Pascal telescoping (the hockey-stick identity) gives $\sum_{k=3}^{40}\binom{k+1}{4} = \binom{42}{5}$.

3. The sum is $3 \cdot 850668 = 2552004$, with remainder 4, displayed as AIME answer 004.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2024-II-Q06] | Part B (Q6-Q10)

Answer: 055

1. There are 2^{a-1} sets with maximum a: include a and choose any subset of $1, \ldots, a-1$. Hence $\sum_{a \in A} 2^{a-1} = 2024$.

2. Binary uniqueness and $2024 = 2^{10} + 2^9 + 2^8 + 2^7 + 2^6 + 2^5 + 2^3$ give $A = \{11, 10, 9, 8, 7, 6, 4\}$. Their sum is 55.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2007-II-Q13] | Part C (Q11-Q15)

Answer: 640

1. If bottom values are $x_0, \ldots, x_{10}$, the top is $\sum_i \binom{10}{i} x_i$. Modulo three the only nonzero binomial weights occur at $i = 0, 1, 9, 10$, all equal one.

2. These four binary entries must sum to zero or three, giving $1 + 4 = 5$ choices. The seven others are free, so total $5 \cdot 2^7 = 640$.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2017-I-Q12] | Part C (Q11-Q15)

Answer: 252

1. The number1 is forbidden by $1 \cdot 1 = 1$. If2 and3 are both absent, every subset of $\{4, \ldots, 10\}$ works because products exceed10. This contributes128, including the empty set.

2. If3 is the least element, exclude9 and choose freely from $4, 5, 6, 7, 8, 10$, giving64 sets. If2 is the least, exclude4; the free elements7,8 give a factor4.

3. In the latter case, $\{3, 6, 9\}$ has five valid choices: empty, each singleton, or $\{6, 9\}$. The pair $\{5, 10\}$ has three choices, excluding their joint selection. All remaining product constraints have now been covered, giving $4 \cdot 5 \cdot 3 = 60$ sets.

4. Total $128 + 64 + 60 = 252$.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2023-II-Q11] | Part C (Q11-Q15)

Answer: 081

1. The 32 subsets form 16 complementary pairs. Exactly one member of each pair must be chosen. The full set is chosen and the empty set is not.

2. If a singleton is chosen, the entire family consists of all sets containing its element. This gives five families. Otherwise all four-element sets are chosen, and the only further restriction is that the selected two-element sets pairwise intersect; three-element choices are then fixed by complementation.

3. View two-element sets as edges of K_5. A pairwise-intersecting edge family is empty, a nonempty subfamily of a star, or a triangle. To see completeness, after edges ab, ac are chosen, any edge avoiding a must be bc, which prevents any fourth edge outside that triangle.

4. The counts are 1, $5(2^4 - 1) - 10 = 65$ (single edges counted twice), and $\binom{5}{3} = 10$. Add the five singleton families: $1 + 65 + 10 + 5 = 81$.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2026-I-Q13] | Part C (Q11-Q15)

Answer: 039

1. In the polynomial ring modulo both 503 and $x^{502}-1$, the reduced coefficients of $(1+x)^{10000}$ are precisely S_r. The prime-binomial identity gives $(1+x)^{503}=1+x^{503}=1+x$ in this ring.

2. As $10000=19\cdot 503+443$, reduce the power to $(1+x)^{462}$. Its coefficients $\binom{462}{r}$ for $r\leq 462$ are nonzero modulo 503, since none of the factorial factors contains 503.

3. Only exponents $r=463,\ldots,501$ have zero coefficient. Their number is 39.

Source / 来源